CS383 · SOFTWARE ENGINEERING

Midterm 461 Practice

13/10/2024 · 1 hour · 15 marks · unified exam

Muharib

01 About this midterm

The midterm of term 461, dated 13/10/2024 (10/04/1446 AH). A unified exam, one hour long, marked out of 15. The questions are copied from photos of a solved paper.

Question	1	2	3	4	5	Total
Marks	3	2	3	2	5	**15**

About the solutions

The answers come from the published solved paper, and I checked them against the slides. One item was not answered on the paper (Q1 item 6), and I answered it myself. The mark for Q1 is inferred from the marks table on the paper.

02 Questions

Q1 MCQ · 3 M

Choose the correct answer, then fill up the table with the correct choice:

1. What does product quality depend on?

   a. Development process b. Developer skills c. Project size d. All of the above

2. What is one of the process attributes that focuses on how easy it is to understand the process definition?

   a. Standardization b. Understandability c. Visibility d. Measurability

3. Project schedule chart include:

   a. tasks b. people involved for each task c. milestones d. all of them

4. Project manager should consider four factors in people management include:

   a. Consistence, Respect, Loyalty, Honesty

   b. Consistence, Respect, Speed, Experience

   c. Consistence, Experience, Inclusion, Honesty

   d. Consistence, Respect, Inclusion, Honesty

5. Risk management involves anticipating risks that may affect:

   a. the project schedule b. the quality of project c. a and b d. none of them

6. Which of the following is not considered as a risk in project management?

   a. Specification delays b. Product competition c. Testing d. Hardware delays

Q2

ETHICS · 2 M

According to the 8 principles of the IEEE Code of Ethics (PUBLIC, CLIENT AND EMPLOYER, PRODUCT, JUDGMENT, MANAGEMENT, PROFESSION, COLLEAGUES, SELF), identify the type of ethics reflected in the following descriptions:

SEEPP ethic example	Ethics Principle
Approve software only if they believe that it is safe for use, meets specifications, passes appropriate tests	
Ensure proper and achievable goals and objectives for any project	
Assist colleagues in being fully aware of current standard work practices	
Improve their ability to produce accurate, informative, and well-written documentation	

Q3 PLAN-DRIVEN OR AGILE · 3 M

For each concern, determine whether the Plan Driven or Agile method is the most suitable choice:

Concern	Plan Driven	Agile
1- Emphasis on cost estimation before starting		
2- Flexibility in handling changing requirements		
3- Use of detailed risk management		
4- Incremental delivery of features		
5- Long periods of planning and analysis		
6- Continuous integration and testing		

Q4 FOUR ACTIVITIES · 2 M

What are the 4 key software engineering activities. The activities must be given in a logical order to ensure the successful delivery of software products:

Q5 FR AND NFR · 5 M

After reading the following system description determine the functional and non-functional requirements:

A company hires out tools and equipment (for example drills, power saws, cement mixers, ladders, scaffolding) to customers and requires a computerized system to record details of bookings. Equipment may be booked in advance, or customers may appear at the reception desk and ask if there is an item available for immediate hire. When dealing with a booking or allocating an available item to a customer, the receptionist must check whether the customer has previously hired equipment from the company or is a new customer. For a new customer, the receptionist must enter the customer's details. Otherwise, the receptionist must retrieve the existing customer's record and update any details if necessary. Each page must load within 5 seconds. The minimum period of hire is one day, and all hires are made for several complete days. The return of an item at the end of the hire period is recorded by the receptionist, or by a technician if the reception desk is closed. The system should be available at working hours. The manager of the company requires a summary of the status of all equipment at the beginning of each day, giving details of items out on hire, items booked and items that will be available for hire that day.

Functional Requirements: 7 lines · Nonfunctional Requirements: 3 lines